Student table

Sid	Sname	Dept	Year	Age	CGPA
101	Alice	CSE	3	22	8.5
102	Bob	CSE	2	20	7.8
103	Charlie	ECE	3	23	9.0
104	Diana	ECE	1	19	6.5
105	Eve	Mech	3	24	7.2
106	Frank	CSE	1	18	8.0
107	Grace	ECE	2	21	8.2
108	Harry	Mech	1	19	7.5

Course table

Cid	Cname	Dept	Credits	Instructor	Semester
201	Database	CSE	4	Prof.A	Fall
202	Algorithms	CSE	3	Prof.B	Spring
203	Circuits	ECE	4	Prof.C	Fall
204	Signals	ECE	3	Prof.D	Spring
205	Thermodynamics	Mech	4	Prof.E	Fall

206	Data Structures	CSE	4	Prof.A	Spring
207	Digital Systems	ECE	4	Prof.C	Spring
208	Fluid Mechanics	Mech	3	Prof.F	Fall

Exercise 1 (Selection + Projection + Condition)

List Sname, Age, CGPA of CSE students in 3rd year with CGPA > 8.0.

```
SELECT Sname, Age, CGPA
FROM Student
WHERE Dept = 'CSE' AND Year = 3 AND CGPA > 8.0;
```

Result:

Sname	Age	CGPA
Alice	22	8.5

Exercise 2 (Projection + DISTINCT)

List all distinct combinations of (Dept, Semester) for courses taught in Fall semester.

```
SELECT DISTINCT Dept, Semester
FROM Course
WHERE Semester = 'Fall';
```

Result:

Dept	Semester

CSE	Fall
ECE	Fall
Mech	Fall

Exercise 3 (Selection + Multiple Conditions)

Find students whose Age > 21 OR CGPA >= 9.0; show Sid, Sname, Dept.

```
SELECT Sid, Sname, Dept
FROM Student
WHERE Age > 21 OR CGPA >= 9.0;
```

Result:

Sid	Sname	Dept
101	Alice	CSE
103	Charlie	ECE
105	Eve	Mech

Exercise 4 (Set UNION)

List all Sids from CSE students UNION all Cids from ECE courses.

```
SELECT Sid AS ID FROM Student WHERE Dept = 'CSE'
UNION
SELECT Cid AS ID FROM Course WHERE Dept = 'ECE';
```

Result:

ID
101
102
106
203
204
207

Exercise 5 (Set INTERSECT)

Find IDs (Sids or Cids) that appear in both CSE students and CSE courses.

```
SELECT Sid AS ID FROM Student WHERE Dept = 'CSE'
INTERSECT
SELECT Cid AS ID FROM Course WHERE Dept = 'CSE';
```

Result:

ID
201
202
206

(No overlap in this data, but shows structure)

Exercise 6 (Set Difference: EXCEPT)

List Sids of students who are NOT in 1st year, minus those in Mech.

```
SELECT Sid FROM Student WHERE Year != 1
EXCEPT
SELECT Sid FROM Student WHERE Dept = 'Mech';
```

Result:

Sid
101
102
103
107

Exercise 7 (Cartesian Product + Non- Equi Filter)

Using explicit CROSS JOIN, list (Sname, Cname) pairs where student's Year < course's Credits.

```
SELECT Sname, Cname
FROM Student CROSS JOIN Course
WHERE Student.Year < Course.Credits;
```

Result (many rows):

Sname	Cname	
Diana	Database	*(1 < 4)*
Diana	Algorithms	*(1 < 3)*
Diana	Circuits	*(1 < 4)*
Diana	Signals	*(1 < 3)*
Diana	Thermodynamics	*(1 < 4)*
Diana	Data Structures	*(1 < 4)*
Dian	Digital	*(1 < 4)*

a	Systems	
Diana	Fluid Mechanics	*(1 < 3)*
Frank	Database	*(1 < 4)*
Frank	Algorithms	*(1 < 3)*
...	...	

(Shows 1st year students pair with almost all courses since 1 < 3 or 4)

SQL to Create and Populate Tables

```
-- Student table
CREATE TABLE Student (
    Sid INT PRIMARY KEY, Sname VARCHAR(20), Dept VARCHAR(10),
    Year INT, Age INT, CGPA DECIMAL(3,1)
);

INSERT INTO Student VALUES
(101,'Alice','CSE',3,22,8.5), (102,'Bob','CSE',2,20,7.8),
(103,'Charlie','ECE',3,23,9.0), (104,'Diana','ECE',1,19,6.5),
(105,'Eve','Mech',3,24,7.2), (106,'Frank','CSE',1,18,8.0),
(107,'Grace','ECE',2,21,8.2), (108,'Harry','Mech',1,19,7.5);

-- Course table
CREATE TABLE Course (
    Cid INT PRIMARY KEY, Cname VARCHAR(20), Dept VARCHAR(10),
    Credits INT, Instructor VARCHAR(10), Semester VARCHAR(10)
);

INSERT INTO Course VALUES
(201,'Database','CSE',4,'Prof.A','Fall'),
(202,'Algorithms','CSE',3,'Prof.B','Spring'),
(203,'Circuits','ECE',4,'Prof.C','Fall'),
(204,'Signals','ECE',3,'Prof.D','Spring'),
(205,'Thermodynamics','Mech',4,'Prof.E','Fall'),
(206,'Data Structures','CSE',4,'Prof.A','Spring'),
(207,'Digital Systems','ECE',4,'Prof.C','Spring'),
```

```
(208,'Fluid Mechanics','Mech',3,'Prof.F','Fall');
```